```sql
-- Q1 display all employess
select*from employees;

-- Q2 display emp name and salary
select employee_name, salary from employees;

-- Q3
select employee_name , salary from employees
where salary>60000;

-- Q4
select employee_name , city from employees
where city="hyderabad";

-- Q5
select employee_name , salary from employees
where salary between 50000 and 70000;

-- Q6
select employee_name , city from employees
where city="hyderabad"
or city="bangalore";

-- Q7
select employee_name from employees
where employee_name like "a%";

-- Q8
select employee_name from employees
where employee_name like "%e";

-- Q9
select employee_name , salary from employees
order by salary desc;

-- Q10
select employee_name ,salary from employees
order by salary desc
limit 3;

-- Q11
select count(*) from employees;

-- Q12
select sum(salary) from employees;

-- Q13
select avg(salary) from employees;

-- Q14
select max(salary) from employees;

-- Q15
select min(salary) from employees;

-- Q16
select count(employee_id) from employees
where city = "hyderabad";

-- Q17
select sum(salary)as total_salary from employees
where city = "mumbai";

-- Q18
select avg(salary)as avg_salary from employees
where city = "bangalore";

-- Q19
select count(*) from employees
group by department_id ;

select department_name,
(select count(*) from employees e
where e.department_id = d.department_id)
as emp_count from departments d;

select * from employees as e
inner join departments as d
on e.department_id = d.department_id;

select department_name as d,
count(employee_id) as emp_count from employees as e
inner join departments as d
on e.department_id = d.department_id
group by d.department_name;
```

```sql
-- Q20
select department_name,
avg(salary) as avg_salary from employees as e
inner join departments as d
on e.department_id = d.department_id
group by d.department_name;
```

```sql
-- Q21
select department_name,
min(salary) as min_salary from employees as e
inner join departments as d
on e.department_id = d.department_id
group by d.department_name;
```

```sql
-- Q22
select department_name,
max(salary) as max_salary from employees as e
inner join departments as d
on e.department_id = d.department_id
group by d.department_name;
```

```sql
-- Q23
select department_name,
sum(salary) as total_salary from employees as e
inner join departments as d
on e.department_id = d.department_id
group by d.department_name;
```

```sql
-- Q24
select city,
count(employee_id) from employees
group by city;
```

```sql
-- Q25
select city,
avg(salary) as avg_salary from employees as e
inner join departments as d
on e.department_id = d.department_id
group by e.city;
```

```sql
-- Q26
select city,
count(employee_id) as emp_count from employees
group by city having count(employee_id)>2;
```

```sql
-- Q27
select department_name,
avg(salary) as avg_salary from employees as e
inner join departments as d
on e.department_id = d.department_id
group by d.department_name
having avg(salary)>60000;

select department_name from departments;
```

```sql
-- Q28
select department_name,
sum(salary) as total_salary from employees as e
inner join departments as d
on e.department_id = d.department_id
group by d.department_name
having sum(salary)>150000;
```

```sql
-- Q29
create table student (
student_id int primary key,
student_name varchar(30) not null,
email varchar(30) unique,
age int check(age>=18),city varchar(10) default "hyderabad");

select*from student;
```

```sql
-- Q30
insert into student value
(1001,"Mohsin","mohsin@gmail.com",23,"nanded");
select*from student;

insert into student value
(1002,"nasir","nasir@gmail.com",24,"");
select*from student;
insert into student value
(1003,"sami","sami@gmail.com",23,);
select*from student;
```

```sql
-- Q31
insert into student value
(1004,"nasir","nasir@gmail.com",24,"srinagr")
(1004,"hussain","hussain@gmail.com",25,"gulmarg");

select*from student;

insert into student value
(1003,"sami","sami@gmail.com",23,"hydrabad");
```

```sql
-- Q32
insert into student value
(1004,"aamir@gmail.com",25); -- column count doesnt match
select*from student;
```

```sql
-- Q33
insert into student value
(1004,"tayyab","tayyab@gmail.com",15,"pune"); -- check constraints student_check violated
```

```sql
-- Q35
select e.employee_name,d.department_name
from employees as e
inner join departments as d
on e.department_id = d.department_id;
```

```sql
-- Q36
select employee_name,salary,department_name,location
from employees as e
inner join departments as d
on e.department_id = d.department_id;
```

```sql
-- Q37
select employee_name,department_name
from employees as e
left join departments as d
on e.department_id = d.department_id;
```

```sql
-- Q38
select employee_name,department_name
from employees as e
right join departments as d
on e.department_id = d.department_id;
```

```sql
-- Q39
select employee_name
from employees as e
inner join departments as d
on e.department_id = d.department_id
where d.department_name = "Data Science";
```

```sql
-- Q40
select employee_name
from employees as e
inner join departments as d
on e.department_id = d.department_id
where e.city = "Hyderabad";
```

```sql
-- Q41
select department_name,
count(employee_id)as emp_count
from departments as d
left join employees as e
on d.department_id = e.department_id
group by d.department_id, d.department_name ;
```

```sql
-- Q42
select department_name,
avg(salary) as avg_salary
from employees as e
inner join departments as d
on e.department_id = d.department_id
group by d.department_id, d.department_name;
```

```sql
-- Q43
select * from employees
where salary> (select avg(salary) from employees
);
```

```sql
-- Q44
select * from employees
where salary= (select max(salary) from employees
);
```

```
-- Q45
select * from employees
where salary = (select salary from employees
where employee_name = "alice"
);
```

```
-- Q46
select * from employees
where salary > (select salary from employees
where employee_name = "alice"
);
```

```
-- Q47
select * from employees as e
where salary > (select avg (salary) from employees as e2
where e.department_id= department_id
);
```

```
-- Q48
select department_name,
avg(salary)as avg_salary from employees as e
inner join departments as d
on e.department_id = d.department_id
group by d.department_id, d.department_name
order by avg_salary desc;
```

```
-- Q49
select department_name,
count(employee_id) as emp_count,
avg(salary),
max(salary),
min(salary),
sum(salary) as total_salary
from employees as e
left join departments as d
on e.department_id = d.department_id
group by d.department_id, d.department_name;
```

```
-- Q50
select employee_name,salary,city,department_name from employees as e
inner join departments as d
on e.department_id = d.department_id
where salary> 55000
and city in ("hyderabad", "bangalore")
and salary > (
select avg (salary) from employees
where department_id = e.department_id
);
```